

CSCE 676 DATA MINING · MODULE 09 · LECTURE NOTES

Recommendation

The worked system, and its specific traps.

MODULE OUTCOMES

- Explain collaborative and content-based approaches.
- Explain matrix factorisation and the implicit feedback problem.
- Explain cold start and popularity bias.
- Explain why offline evaluation misleads here.
- Build and evaluate a recommender honestly.

1 Two approaches

	Collaborative filtering	Content-based
What it uses	Who liked what — the interaction matrix only.	Item attributes — text, metadata, categories, features.
Its strength	Finds surprising connections nobody could specify — see the callout.	Works for a brand-new item immediately, with no interactions at all.
Cold start	Fails for new users <i>and</i> new items.	Fails for new users only — which is half the problem solved.
Its bias	Toward whatever is popular (§3).	Toward whatever is similar — the filter bubble, which is a different failure.
What it needs	A great deal of interaction data.	Good item features, which someone has to produce and maintain.

So real systems are hybrid — content-based for the cold start and collaborative once there is enough interaction data, with the switch between them being an explicit design decision rather than an emergent one. The complementary-failure framing explains why hybrids dominate practice even though the research literature treats the two families separately.

COLLABORATIVE FILTERING'S STRENGTH IS FINDING CONNECTIONS NOBODY COULD SPECIFY

It requires no understanding of the items whatsoever. If users who liked A also liked B, that is sufficient — and **the connection may have no describable feature behind it at all**, which is the whole appeal.

Which is genuinely powerful, because the features that predict taste are frequently not the features anybody would have thought to record — and a content-based system can only use features somebody chose to capture.

And it is the same distributional argument as CSCE 638 Module 04's — meaning from co-occurrence, applied to preferences instead of to words, with the same mathematics (matrix factorisation of a co-occurrence structure) underneath.

With the same limitation, too: it cannot say anything at all about an item nobody has interacted with — which is the cold start problem (§3) and is **structural rather than incidental**, exactly as a word absent from the corpus has no embedding.

2 Matrix factorisation

$$r_{ui} = \mu + b_u + b_i + p_u \cdot q_i$$

- **A global mean, a user bias, an item bias, and a latent factor interaction. The bias terms matter considerably more than people expect** — modelling that some users rate generously and some items are generally liked captures a large share of the variance before any factors are involved.
- **The objective is summed over the *observed* entries only**, with regularisation on the factors — **which is precisely what distinguishes this from the SVD of Module 05**.
- **So it is not an SVD**, despite being universally described as one. **SVD requires a complete matrix; this is a regularised fit to the observed entries**, solved by stochastic gradient descent or by alternating least squares. **The not-actually-SVD point is a common confusion worth clearing up**, because it explains why the Eckart–Young guarantee does not apply here.
- **And the regularisation is doing real work**, because the observed entries are a tiny and non-random fraction of the matrix — which is §3's feedback loop arriving as a fitting problem.

IMPLICIT FEEDBACK CHANGES THE PROBLEM, BECAUSE A MISSING ENTRY IS AMBIGUOUS

Explicit ratings are rare, noisy, and unrepresentative; clicks, views, plays, and purchases are abundant — so **real systems use implicit feedback almost exclusively**, and the explicit-rating literature describes a setting that mostly no longer exists.

But an absent interaction is not a negative rating. The user may dislike the item, or may simply never have seen it — and **the two are not distinguishable from the logged data**, which is the central difficulty.

So the standard approaches are either to treat all unobserved pairs as weak negatives with a confidence weight — weighting observed interactions by their strength — **or to sample negatives and optimise a ranking objective directly**, as in Bayesian personalised ranking.

And it means the objective changes fundamentally: you are producing a ranking rather than predicting a value — so **squared error on held-out ratings is measuring the wrong thing**, which is §4's first metric point and is still a common mistake.

3 Cold start and bias

- **New user cold start. No history, so no collaborative signal exists** — addressed by onboarding questions, demographic priors, or popular defaults, **all of which are weak** and all of which annoy the user to some degree.
- **New item cold start. No interactions, so it cannot be recommended, so it receives no interactions** — a **self-reinforcing feedback loop that content features break** and that pure collaborative filtering cannot escape.
- **Popularity bias. Popular items have more interaction data, so they are predicted more confidently, so they are recommended more, so they accumulate more data** — **which means the system amplifies the popularity distribution rather than merely reflecting it.**
- **And the feedback loop, which is the general case. Your training data is the output of your previous recommender, so the model learns from its own past choices** rather than from independent evidence about preferences.
- **Which makes the logged data not a sample of user preferences but a sample of what you chose to show. The feedback loop is the deepest of these problems** — **it means your offline evaluation data was generated by the very policy you are trying to improve on**, which is §4's subject and is not fixable by a better model.

4 Evaluation

OFFLINE EVALUATION ON LOGGED DATA SYSTEMATICALLY FAVOURS WHAT THE OLD SYSTEM DID

You can only evaluate recommendations the user actually saw — there is no record of what they would have thought of anything else — so **a new system that recommends something genuinely good but previously unshown scores zero for it**, and **a system that imitates the old one scores well.**

Which biases offline evaluation systematically toward the status quo, and is why offline gains frequently fail to appear in an online A/B test — a mismatch that is routine and widely underreported.

And the replication findings here are stark: several years of reported neural recommender improvements did not survive comparison against properly tuned simple baselines — a result reproduced across multiple surveys, and one of the cleanest examples of Module 13's problem in any field.

So: use ranking metrics at a realistic k, report popularity bias and catalogue coverage, correct for the logging policy where you can (inverse propensity weighting), and test online if the decision matters — **which is the honest programme**, and each step is achievable.

- **Ranking metrics at a realistic k** — precision@k, recall@k, and NDCG, where k is the number of items you actually show — **not RMSE, which measures the wrong thing for a ranked list (§2).**
- **Coverage: what fraction of the catalogue is ever recommended to anybody? A system recommending two hundred items out of a million is doing something you should know about**, and no accuracy metric reveals it.

- **Popularity bias: plot the recommended items' popularity distribution against the catalogue's**, which quantifies §3's **amplification** and turns a concern into a number.
- **Diversity and novelty within a single list**, since **a list of ten near-identical items is worse than its per-item scores suggest** — the list is the unit the user experiences, and per-item metrics do not see it.
- **And a properly tuned simple baseline — popularity, and item-item nearest neighbours** — which is **the requirement most often skipped** and the one the replication work turned on. **Item-item nearest neighbours, properly tuned, is the baseline that has embarrassed a decade of published work**, so **run it first** and know what you have to beat.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Mining of Massive Datasets, chapter 9 (free)	§1 and §2, with the collaborative filtering and latent factor material done cleanly.
Hu, Koren & Volinsky — Collaborative Filtering for Implicit Feedback (free)	§2's callout in the original — the confidence weighting approach, and why the problem differs.
Dacrema, Cremonesi & Jannach — Are We Really Making Much Progress? (free)	§4's replication finding — the paper that forced the field to tune its baselines. Read it before Module 13.
Chaney, Stewart & Engelhardt — How algorithmic confounding biases recommendation (free)	§3's feedback loop, simulated — what happens to the data over time under a deployed recommender.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Build a content-based and a collaborative recommender** on the same data, and compare their cold-start behaviour.
- 2 **Implement biased matrix factorisation** and measure how much the bias terms alone achieve.
- 3 **Explain why the objective is not an SVD**, in your own words.
- 4 **Convert explicit ratings to implicit feedback** and refit with a ranking objective.
- 5 **Measure the recommended items' popularity distribution** against the catalogue's.
- 6 **Compute catalogue coverage** for your system.

- 7 **Simulate the feedback loop:** retrain on your own recommendations' outcomes for several rounds and watch coverage.
- 8 **Compute RMSE and NDCG@10** for the same system and compare the rankings of two models.
- 9 **Tune item-item nearest neighbours properly** and compare against your best model.
- 10 **Report the comparison honestly**, whichever way it came out.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Contrast the two approaches on five dimensions.
- 2 Why are real systems hybrid?
- 3 What is collaborative filtering's distinctive strength, and which other course shares the argument?
- 4 Give the factorisation model and say why the bias terms matter.
- 5 Why is it not an SVD?
- 6 Why is a missing implicit entry ambiguous, and what are the two responses?
- 7 Name four structural problems, and which is deepest.
- 8 Why does offline evaluation favour the status quo?
- 9 What did the replication work find?
- 10 Give five things to measure, and the baseline to run first.